

Dioxins, dibenzofurans, dioxin-like PCBs, and DDE in U.S. fast food, 1995.



Food, especially dairy products, meat, and fish, is the primary source of environmental exposure to dioxins in the general population. Little data exists on dioxin levels in the popular and widely consumed "fast foods". Data presented in a previously published pilot study was limited to measuring only the levels of dioxins and dibenzofurans in three types of U.S. fast food. This study adds to the previous paper by presenting data, in addition to dioxins and dibenzofurans, on the closely related dioxin-like polychlorinated biphenyls (PCBs), and the persistent metabolite of DDT, 1,1-dichloro-2,2-bis (p-chlorophenyl) ethylene (DDE), in four types of popular U.S. fast food. These include McDonald's Big Mac Hamburger, Pizza Hut's Personal Pan Pizza Supreme, Kentucky Fried Chicken (KFC) three piece original recipe mixed dark and white meat luncheon package, and Häagen-Daz chocolate-chocolate chip ice cream. Dioxin plus dibenzofuran dioxin toxic equivalents (TEQ) ranged from 0.03 to 0.28 TEQ pg/g wet or whole weight for the Big Mac, from 0.03 to 0.29 for the Pizza, from 0.01 to 0.31 for the KFC, and from 0.03 to 0.49 TEQ pg/g for the ice cream. Daily TEQ consumption per kilogram body weight (kg/BW), assuming an average 65 kg adult and a 20 kg child, from one serving of each of these fast food ranged between 0.046 and 1.556 pg/kg in adults whereas in children the values were between 0.15 and 5.05 pg/kg. Total measured PCDD/Fs in the Big Mac, Personal Pan Pizza, KFC, and the Häagen-Daz ice cream varied from 0.58 to 9.31 pg/g. Measured DDE levels in the fast foods ranged from 180 to 3170 pg/g. Total mono-ortho PCB levels ranged up to 500 pg/g or 1.28 TEQ pg/g for the KFC and for di-ortho PCBs up to 740 pg/g or 0.014 TEQ pg/g for the pizza sample. Total PCB values in the four samples ranged up to 1170 pg/g or 1.29 TEQ pg/g for the chicken sample.